C.V. RAMAN GLOBAL UNIVERSITY, BHUBANESWAR

SUBJECT: MATHEMATICS-II (MTH21102)

BTECH (2^{nd} Sem) [COMMON TO ALL BRANCHES]

	Unit – IV [Ordinary Differential Equations Of Second Order]
1.	Assuming one solution $y_1 = x^{-1/2}\cos x$, find another solution of the equation $x^2y'' + xy' + (x^2 - 1/4)y = 0$.
2.	Assuming one solution $y_1 = \sin x/x$, find another solution of the equation $xy'' + 2y' + xy = 0$.
3.	Assuming one solution $y_1 = x$, find another solution of the equation $(1 - x^2)y'' - 2xy' + 2y = 0$.
4.	Find the general solution of the following ordinary differential equations. (i) $(D^3 + 1)y = e^x\cos x + \sin 3x$. (ii)) $(D^2 + 2D + 2)y = 4e^{-x}\sec^3 x$. (iii) $y'' - 2y' + 5y = 5x^3 - 6x^2 + 6x$ (iv) $(D^2 + 1)y = -x^{-2} + \ln \pi x$ (v) $(D^2 - 2D + 17)y = e^{-x}\sin 4x$. (vi) $(D^2 + 4D + 4)y = e^{-2x}/x^2$, $y(1) = 1/e^2$, $y'(1) = -2/e^2$, (vii) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 2x^2 + e^x + 2xe^x + 4e^{3x}$ (viii) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x\sin^{-1}x$. (ix) $\frac{d^2y}{dx^2} + 4y = 12x^2 - 16\,x\cos 2x$
5.	Solve the following initial value problems. (i) $x^2y'' - 2xy' + 2y = 0, y(1) = 1.5, y'(1) = 1$. (ii) $4x^2y'' + 24xy' + 25y = 0, y(1) = 2, y'(1) = -6$. (iii) $x^2y'' + xy' + 9y = 0, y(1) = 2, y'(1) = 0$. (iv) $(x^2D^2 - 3xD + 4)y, y(1) = 0, y'(1) = 3$. (v) $(x^2D^2 + 3xD + 1)y, y(1) = 3, y'(1) = -4$.
6.	Find the steady state oscillation of the mass – spring system governed by the given equation. Show the details of your calculation. (i) $(D^2 + 4D + 3)y = \cos t + \frac{1}{3}\cos 3t$. (ii) $(D^2 + D + 1)y = e^t$
7.	Find the transient motion of the mass – spring system governed by the given equation. Show the details of your calculation. (i) $(D^2 + 2D + 2)y = 170\sin 4t$. (ii) $(D^2 + 6D + 9)y = 25\sin t$
8.	Find the motion of the mass – spring system corresponding to the given equation and initial conditions. Stat the time when the solution practically reaches the steady state. (i) $(D^2 + 4)y = \sin t + \frac{1}{3}\sin 3t + \frac{1}{5}\sin 5t$, $y(0) = 1$, $y'(0) = \frac{3}{25}$. (ii) (ii) $(D^2 + 8D + 17)y = 474.5\sin 0.5t$, $y(0) = -5.4$, $y'(0) = 9.4$.
9.	Find the steady state current in the RLC circuit with (i) R = 8 Ohms, L = 2henrys, C = 0.1 farad, E = 160cos5t volts. (ii) R = 2Ohms, L = 1henrys, C = 0.5 farad, E = 50sint volts.
10.	Find the current in the RLC circuit with (i) R = 40 Ohms, L = 0.5henry, C = 1/750 farad, E = 25cos100t volts. (ii) R = 1Ohms, L = 0.1henrys, C = 1/340 farad, E = e^{-t}(169.9 sint -160.1 cost) volts.
11.	Find the current of the of the following RLC circuit with zero initial current and charge (i) R = 8 Ohms, L = 10henrys, C = 0.004 farad, E = 240.5sin10t volts. (ii) R = 3hms, L = 0.5henrys, C = 0.08 farad, E = 12cos5t volts.
12.	Find the current of the of the following LC circuit with zero initial current and charge (i) L = 10henrys, C = 0.1 farad, E = 10 t volts.

	(ii) L = 2henrys, C = 0.005 farad, E = 220 sin4t volts.
13.	An 8lb weight is placed at one end of a spring suspended from the ceiling. The weight raised to 5 inches above the equilibrium position and left free. Assuming the spring constant 12lb/ft, find the equation of motion, displacement function *x(t)*, amplitude, period, frequency and maximum velocity.
14.	A 2lb weight is suspended from one end of a spring stretches it to 6 inches. A velocity of 5ft/sec upwards is imparted to the weight at its equilibrium position. Suppose a damping force βv acts on the weight. Here $0 < \beta < 1$ and $v = \dot{x}$ = velocity. (a) Determine the position and velocity of the weight at any time. (b) Express the displacement function *x(t)* in the amplitude phase form, (c) Find the amplitude, period, frequency, maximum velocity, (d) Determine the values of β for which the system is critically damped, over –damped or oscillatory, (e) Discuss the case for $\beta = 0.6$.
15.	Solve the following boundary value problems. (i) $y'' + 4y = 0, y(0) = 3, y\left(\frac{\pi}{2}\right) = 0.$ (ii) $y'' - 25y = 0, y(-2) = cosh10, y(2) = cosh10.$ (iii) $y'' + 2y' + 2y = 0, y(0) = 1, y\left(\frac{\pi}{2}\right) = 0.$ (iv) $3y'' - 8y' - 3y = 0, y(-3) = 1, y(3) = 1/e^2$

SHORT QUESTIONS

1.	Solve $y'' + 4y = 0, y(0) = 3, y\left(\frac{\pi}{2}\right) = 0.$
2.	Are the following functions linearly independent or dependent on the given interval (a) $x\|x\|, x^2, 0 \leq x \leq 1$ (b) $sin^2 x, sin\, x^2, 0 < x < \sqrt{\pi}.$
3.	Find the general solution of $x^2y'' + xy' + y = 0.$
4.	Define a linear differential equation of second order and give one example.
5.	Find the solution of the differential equation $\frac{d^2y}{dt^2} + 4y = 0.$
6.	Predict one of the solutions to the differential equation $y'' + xy' - y = 0.$
7.	Find a particular solution of $y'' + y = x^3.$
8.	Solve $(D^2 - 2D + 4)y = 0$.
9.	Solve $4y'' - 25y = 0; y(0) = 0;\ y'(0) = -5.$
10.	Solve $y'' + 4y' + 4y = 0; y(0) = 1;\ y'(0) = 1.$
11.	Solve $16y'' - \pi^2 y = 0.$
12.	Solve $(D^2 + \pi(\pi - 1)D - \pi^3)y = 0.$
13.	Find the Wronskian of the following pairs of functions (i) $e^{-\frac{ax}{2}}cos3x, e^{-\frac{ax}{2}}sin3x$ (ii) $x^4, x^4 lnx$
14.	Find the particular integral of the differential equation $(D^2 - 4D + 4)y = e^x$
15.	Find the particular integral of the differential equation $(D^2 - 4D + 4)y = x$
16.	Find the particular integral of the differential equation $(D^2 + 16)y = cos2x$
17.	Write down the differential equation which governs the current in the circuit with $R = 10\Omega, L = 0.1H, C = 10^{-6}\mu F$ and emf $E = 10\sin 50t$.
18.	Write down the differential equation which governs the current in the *LC* circuit with, $L = 10H, C = 4 \times 10^{-1}F$ and emf $E = 10t\ Volts$.
19.	Find a second order differential equation with $4e^{3x} + 5e^{-3x}$ as its solution.
20.	Find a second order differential equation with $4 sin2x + 7 cos2x$ as its solution.